From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - I

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 02 and 09 to 30)** contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (03 to 08) contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. In a standard meter bridge experiment, to measure the specific resistance of a wire the following data are found:-
Length (L) $= (1 \pm \cdot 01)$m measured by a meter scale
Radius of wire = 1mm $\pm$.01mm. (measured by screw gauge).
Resistance of wire (R) = (5$\pm$.01) Ω
The maximum possible error in the measurement of specific resistance is
(You may use the formula:- $\rho = \frac{\pi r^2}{L} R$)

(A) $\cdot 1 \mu\Omega$m (B) $\cdot 2 \mu\Omega$m
(C) $\cdot 5024 \mu\Omega$m (D) $\cdot 6024 \mu\Omega$m

2. A physical quantity P is related to four measurable quantities as $P = \frac{a^3 b^2}{c^3 d^{4.}}$. If the errors in measurement of a, b, c, d are 0.1%, 0.2%, 0.3%, 0.1% respectively, what is the maximum percentage error in measurement of P.
(A) 1% (B) 2%
(C) 3% (D) 4%

3. When N_0 molecules of a radioactive nuclide are taken at t = 0, the activity reduces to 'A' in time t_0. When $3N_o$ molecules of the same nuclide are taken, activity will become 3Ae in time t equal to :-

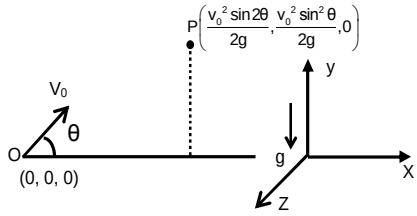
(A) t_0 (B) $t_0 + \frac{1}{\lambda}$

(C) $t_0 - \frac{1}{\lambda}$ (D) None of these

Where; λ is decay constant of the Nuclide

Space for rough work

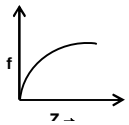
4. A particle of mass m is projected from O with velocity $v_0\cos\theta\hat{i} + v_0\sin\theta\hat{j}$.
At the same instant, another particle of mass 2m is projected from point P with $v_0\hat{k}$. The velocity of 2nd particle w.r.t first particle after time $t = \frac{v_0\sin\theta}{g}$ is

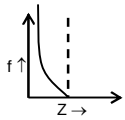


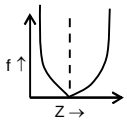
(A) $v_0\hat{k} + gt\hat{j} + v_0\cos\theta\hat{i}$ (B) $v_0\hat{k} - gt\hat{j} + v_0\cos\theta\hat{i}$

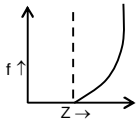
(C) $v_0\hat{k} - v_0\cos\theta\hat{i}$ (D) None of the above

5. The plot of characteristics frequency 'f' in case of X – rays of K_γ wavelength, with the atomic number 'Z' of an element may be best represented by :-

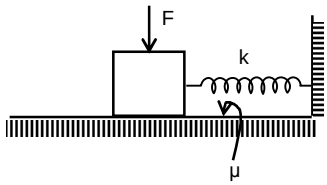
(A)


(B)


(C)


(D)


6. In the position shown, the spring is at its natural length. The block of mass m is given a velocity v_0 towards the vertical support at t = 0. The coefficient of friction between the block and the surface is given by: $\mu = \alpha x$, where α is a positive constant and x is the position of the block from its starting position. The block comes to rest for the first time at x =



(A) $v_0\sqrt{\frac{m}{k+\alpha mg}}$ (B) $v_0\sqrt{\frac{m}{k}}$

(C) $v_0\sqrt{\frac{m}{\alpha mg}}$ (D) none of the above.

Space for rough work

7. Two waves of Intensity I_1 and I_2 interfere. The maximum intensity produced to the minimum intensity are in the ration α. Then I_1/I_2 is equal to

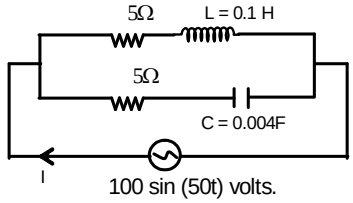
(A) $\frac{\alpha - 1}{\alpha + 1}$, if $I_1 > I_2$

(B) $\frac{1 + \alpha}{\alpha - 1}$, if $I_1 > I_2$

(C) $\left(\frac{1 + \sqrt{\alpha}}{\sqrt{\alpha} - 1}\right)^2$, if $I_1 < I_2$

(D) $\left(\frac{\sqrt{\alpha} - 1}{1 + \sqrt{\alpha}}\right)^2$ if $I_1 > I_2$

8. The current in the circuit at any time t is



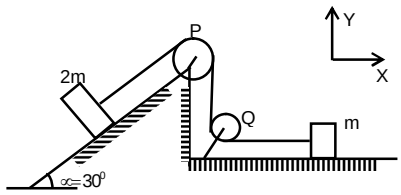
(A) 20ampere.

(B) $20\sin\left[50t + \frac{\pi}{4}\right]$

(C) $20\sin\left[50t\ \frac{-\pi}{4}\right]$

(D) $20\sin(50t)$

9. All the surfaces & pulleys are frictionless in the shown arrangement. Pulleys P and Q are massless. The force applied by clamp on pulley P is



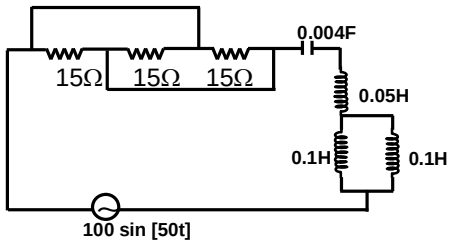
(A) $\frac{mg}{6}\left(-\sqrt{3}\hat{i} - 3\hat{j}\right)$

(B) $\frac{mg}{6}\left(\sqrt{3}\hat{i} + 3\hat{j}\right)$

(C) $\frac{mg}{6}\sqrt{2}$

(D) none of the above.

10. The power factor, ϕ, for the above circuit and the peak current, neglecting the mutual inductance of the coils, is :-



(A) $\frac{\pi}{4}, 10\sqrt{2}$ Ampere

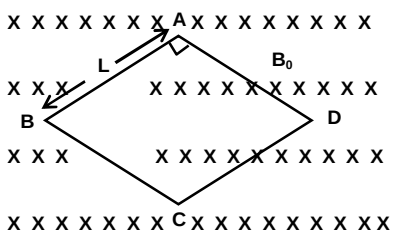
(B) $\frac{-\pi}{4}, 10\sqrt{2}$ Ampere.

(C) 0, 20 Ampere

(D) $\frac{\pi}{4}, 10$ Ampere

Space for rough work

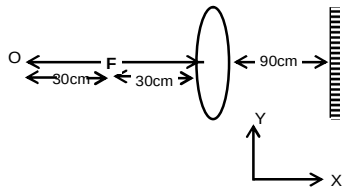
11. A conducting square loop of length 'L' is placed in a magnetic field of Induction $B_0 \cdot$ End 'A' and End 'C' are moved away from each other. So that finally the square loop becomes a straight line. The total charge flowing through the loop, if its resistance is 'R' then :



(A) $\frac{B_0L^2}{\pi R}$ (B) $< \frac{B_0L^2}{R}$

(C) $\frac{B_0L^2}{R}$ (D) $> \frac{B_0L^2}{R}$

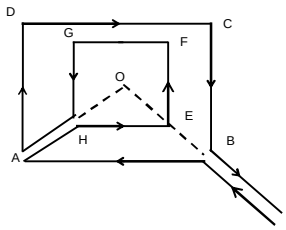
12. A thin biconvex lens of focal length 30cm is Kept 90 cm away from a plane mirror kept parallel to the plane of the lens. A point object is placed on the optical axis of the lens 60cm away from the lens as shown in the figure. Now, if the point object is slowly moved towards the lens with a small velocity, the image of the object in the mirror.



(A) Moves in the + ve x direction (B) Moves in – ve x direction

(C) Does not move (D) Data insufficient

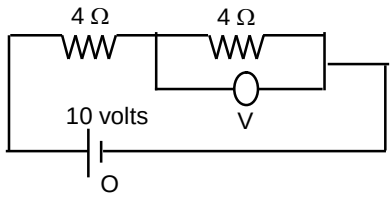
13. ABCD is a square of side 4R and EFGH is a square loop of side 2R. The loop carries a current I. The magnetic field at the centre O of the loop



(A) $\frac{\mu_0 I}{2\pi R} \odot$ (B) $\frac{\mu_0 I}{2\pi R} \otimes$

(C) $\frac{\mu_0 I}{\sqrt{2}\pi R} \otimes$ (D) $\frac{\mu_0 I}{\sqrt{2}\pi R} \odot$

14. In the given circuit, if the voltmeter is ideal its reading is V_1 volts. If the voltmeter has a resistance of 100Ω, the reading is V_2. Then $|V_1 - V_2|$ is equal to:-

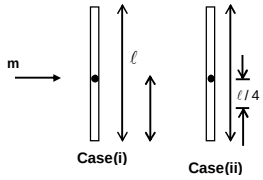


(A) Zero (B) 0.3volts.

(C) 0.2volts (D) 0.1volts

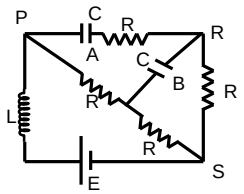
Space for rough work

15. Two identical rods each of mass m and length ℓ are placed on a smooth horizontal surface. In case (1) a particle of mass m moving with V_0 strikes the rod perpendicularly at its centre. In case (ii), another particle of mass m moving with V_0 strikes the rod perpendicularly at $\ell/4$ from the centre. In both the cases, particles come to rest just after collision. The ratio of velocity of center of mass of rod in case (i) to that in case (ii) is



(A) 1:1 (B) 1:2
(C) 2:1 (D) 4:1

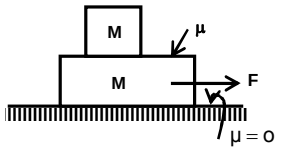
16. The charge on the capacitors A and B in steady state is :-



(A) $\frac{CE}{2}, CE$ (B) $\frac{CE}{2}, \frac{CE}{2}$
(C) $CE, \frac{CE}{2}$ (D) CE, CE

17. In a region, uniform electric field exists as $\vec{E} = \left[10\hat{i} + 10\hat{j}\right] N/C$. If the potential of origin is 0 volts, the potential of point (10m, 10m, 10m) is
(A) 20 volts (B) – 200 volts
(C) 10 volts (D) – 10 volts

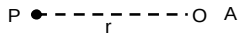
18. For $\mu = \frac{F}{Mg}$, the power applied by force F on the system as a function of time t is



(A) Zero (B) $\frac{F^2}{2M}t$
(C) $\frac{F^2}{M}t$ (D) none of the above

Space for rough work

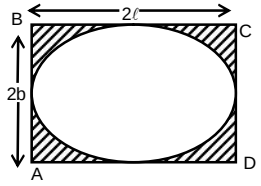
19. There in a small source of heat radiations emitting energy at a constant rate of P watts. At a distance r from the source a small thin disc of mass 'M' surface area 'A' and specific heat capacity 'c' is kept. The time required to raise the temperature of the disc by T Kelvin is



(A) $\frac{mcT}{P}$ (B) $\frac{4\pi r^2 mcT}{PA}$

(C) $\frac{\pi r^2 mcT}{PA}$ (D) data insufficient.

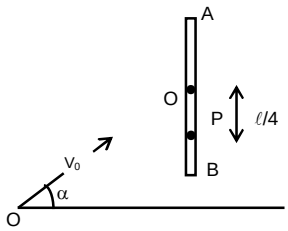
20. There is a thin plate ABCD with an elliptical hole as shown in the figure. The coefficient of linear expansion of the sheet is α. When the temperature of the plate is increased by T K, the area of the hole increases by an amount :-



(A)$\pi\ell b\ \alpha T$ (B) $2\pi\ell b\alpha T$

(C) $3\pi\ell b\ \alpha T$ (D) Data insufficient.

21. A rod of mass M and length ℓ is falling vertically downwards. A particle of m is projected with V_0 at t = 0 at an angle α with horizontal and it strikes the rod perpendicularly at $P(\text{at } t = t_1)$ and comes to rest. The rod strikes the ground at $t = t_2$. The angular velocity of rod about its centre of mass (O)



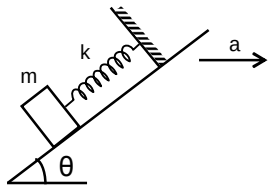
(A) keeps on increasing from t_1 to t_2

(B) keeps on decreasing for t_1 to t_2

(C) remains constant and is non – zero for t = t_1 to t_2

(D) remains constant and is zero for $t = t_1$ to t_2

22. A spring block system (mass = m, spring constant k) is placed on a smooth inclined plane (θ = angle of inclination). The plane is accelerated horizontally with a such that the block does not loose contact with the plane. The time period of small oscillation of the block is



(A) $2\pi\sqrt{\frac{m}{k}}$ (B) $2\pi\sqrt{\frac{m\sin\theta}{k}}$

(C) $2\pi\sqrt{\frac{mg}{ka}}$ (D) none of the above.

Space for rough work

23. A spherical drop of radius r and density n is falling in air with terminal velocity. The density of air is n_0 and its coefficient of viscosity μ. The power developed of by gravity is

(A) $\frac{\pi r^5 n}{\mu}(n-n_0)g^2$ (B) $\frac{\pi r^5}{2\mu}n[n-n_0]$

(C) $\frac{27}{8\mu}r^5 n(n-n_0)g^2$ (D) $\frac{8\pi r^5}{27\mu}n[n-n_0]g^2$

24. A progressive mechanical longitudinal wave is represented by $y = A\sin(\omega t - kx + \pi)$

For $k = 100cm^{-1}$, phase difference between two particles at $x_1 = 2cm$ and $x_2 = 4cm$ at $t = 0$ is

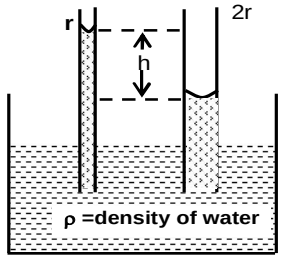
(A) 1rad (B) 2rad

(C) 3rad (D) none of above.

25. In the given arrangement, two capillaries of glass having radii r and 2 r are dipped in pure water of density ρ, the surface tension of water is

(A) $2\rho hgr$ (B) ρhgr

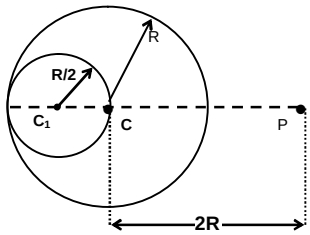
(C) $\frac{\rho hgr}{2}$ (D) None of these



26. A solid uniform sphere of mass M and radius R with centre C is isolated in space. If in the sphere a cavity of radius $\frac{R}{2}$ with centre C_1 is made, then the potential at a point P at a distance of 2R from centre C is

(A) $< -\frac{2GM}{5R}$ (B) $> -\frac{2GM}{5R}$

(C) $= -\frac{2GM}{5R}$ (D) None of these.



Space for rough work

27. Two concentric circles each of radius a (50m< a < 100m) lie in mutually perpendicular planes. A source of sound moves with constant speed on one circle and an observer with same speed on the other circle. If both cross each other simultaneously by at a point, then actual frequency (f_1) of sound and apparent frequency (f_2) received by observer are such that

(A) $f_1 < f_2$ always (B) $f_1 = f_2$ always

(C) $f_1 > f_2$ always (D) none of above

28. A satellite of mass m is orbiting round the earth at a distance of 2R' from the centre of the earth. The amount of energy that must be given to the satellite in order to make its radius 6R must be :-

(A) less than $\frac{GM_em}{3R}$ (B) equal to $\frac{GM_em}{6R}$

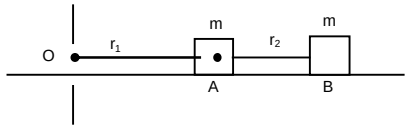
(C) greater than $\frac{GM_em}{4R}$ (D) less than $\frac{GM_em}{4R}$

29. A person throws a ball (A) vertically upwards with 40 m/sec. After 4 sec, he throws another identical ball (B) upwards with same speed. If the collision between balls is head-on perfectly elastic, the speed of ball (A) just after collision is

(A) 0 (B) 10 m/sec

(C) 20 m/sec (D) 40 m/sec

30. Two blocks of mass m each are connected by a massless string while block A is connected to point O by another massless string. Both rotate with constant speed ω on the horizontal surface with O as centre. If T_1 is the tension in OA and T_2 is tension in string AB, then the relation between T_1 & T_2 is:-



(A) $T_1 = T_2$ (B) $T_1 > T_2$

(C) $T_1 < T_2$ (D) data insufficient

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. In a mixture of C_6H_5Cl and $C_6H_5CH_3$. The $\chi_{C_6H_5Cl}$ is 0.5. The maximum % composition of C_6H_5Cl is the mixture is:
   (A) 35% (B) 45%
   (C) 55% (D) 65%

2. A gas under constant temperature was kept to the change in this manner that the density of gas increase twice of its original density so the diffusability of gas will be. (assuming pressure and volume varies remarkably)
   (A) initially decreases and then increases (B) initially increases then decreases
   (C) remain as it is (D) it is negative work done

3. The sum of angular node and angular momentum for ψ_{432} is
   (A) $\frac{\sqrt{6}h+3\pi}{2\pi}$ (B) $\frac{\sqrt{3}h+\pi}{\pi}$
   (C) $\frac{\sqrt{3}h+3\pi}{\pi}$ (D) $\frac{\sqrt{3}h+3\pi}{3\pi}$

4. On heating orthophosphoric acid at 250°C it gives (assuming no further reaction).
   (A) Phosphorus acid (B) Hypophosphoric acid
   (C) Pysophosphoric acid (D) Metaphosphoric acid

5. Inorganic benzene is $B_3N_3H_6$, if all the H are replaced by OH group and all the N by P atom. Then possibility of π bond
   (A) B to O and O to P (B) B to O and P to O
   (C) O to B and O to P (D) O to B and P to O

6. In NaOX, X can not be
   (A) F_2 (B) Cl_2
   (C) Br_2 (D) I_2

Space for rough work

7. $K_2Cr_2O_7$ when reacts with cold conc. H_2SO_4 gives red crystal of

(A) CrO_4^{-2} (B) CrO_3

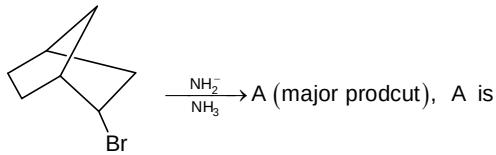
(C) $Cr_2(SO_4)_3$ (D) Cr_2O_3

8. Which of the following statement is incorrect about $\left[Cu(NH_3)_4\right]SO_4$

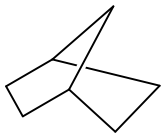
(A) It is paramagnetic in nature (B) the hybridization of the complex is dsp^2

(C) The colour of the complex is blue (D) SO_4^{-2} behaves as ligand

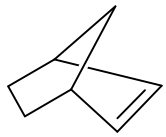
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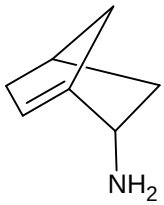
(A)



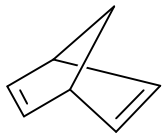
(B)



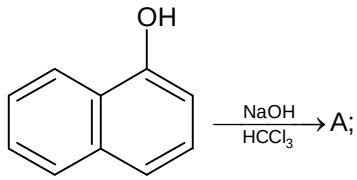
(C)



(D)

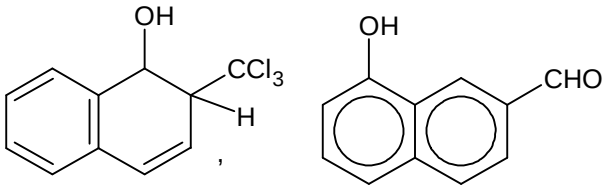


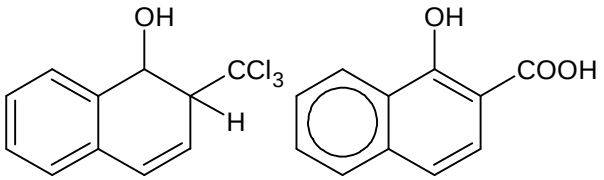
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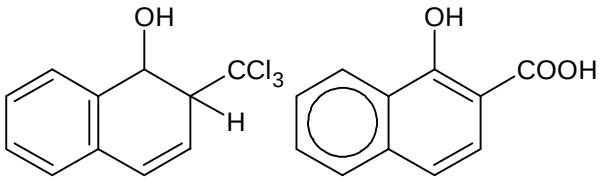
10. 

α – Naphthol

The intermediate of the reaction with the major product is

(A) 

(B) 

(C) 

(D) None of these

11. $Na_2S_2O_3$ when treated with $AgNO_3$ in presence of heat, gives black ppt. of
(A) $Ag_2S_2O_3$ (B) Ag_2O
(C) Ag_2S (D) Na_2S

12. Critical temperature (T_c) for 2 gases X and Y is T_{c1} and T_{c2} respectively, so the radius of X and Y are related as follow of $T_{c1} > T_{c2}$
(A) $r_x > r_y$ (B) $r_y > r_x$
(C) $r_y = r_x$ (D) $r_x + r_y = 0$

13. The coordination number of Wurtzite structure is
(A) 2 : 2 (B) 4 : 4
(C) 6 : 6 (D) 8 : 8

14. When equal conc. of aniline in added to acetone which of the following property of solution will not change
(A) Molarity (B) Boiling point
(C) Volality (D) Vapour pressure

Space for rough work

15. For a consecutive reaction
$A \rightarrow B \rightarrow C$ the $[B]_{max}$ will be

(A) $[B]_{max} = [A]_t \times \left[\frac{K_2}{K_1}\right]^{\frac{K_2}{K_1-K_2}}$

(B) $[B]_{max} = [A]_0 \times \left[\frac{K_2}{K_1}\right]^{\frac{K_2}{K_1-K_2}}$

(C) $[B]_{max} = [A]_t \times \left[\frac{K_1}{K2}\right]^{\frac{K_2}{K_1-K_2}}$

(D) $[B]_{max} = [A]_0 \times \left[\frac{K_1}{K_2}\right]^{\frac{K_2}{K_1-K_2}}$

16. In the reaction of solubility of $\left[Fe(H_2O)_6\right]^{3+} + H_2O \longrightarrow H_3O^+ + \left[Fe(H_2O)_5\,OH\right]^{+2}$
The complex formation takes place if

(A) high concentration of $\left[Fe(H_2O)_5\,OH\right]^{+2}$ is taken

(B) Ksp is positive

(C) low concentration of $\left[Fe(H_2O)_5\,OH\right]^{+2}$ is taken

(D) Ksp is negative

17. The pH of the solution of 0.1 M is acetic acid and 0.1 M benzoic acid is
(if $K_{a_{C_6H_5COOH}} = 6.5 \times 10^{-5}$, $K_{a_{CH_3COOH}} = 1.8 \times 10^{-5}$)

(A) 3.5
(B) 2.5
(C) 1.5
(D) 0.5

18. $A^{+a} + ae^- \rightleftharpoons A \qquad E_1^0$

$A^{b+} + be^- \rightleftharpoons A \qquad E_2^0$

For simultaneous reactions electrode potential is

(A) $E_3 = \frac{aE_1^0 + bE_2^0}{(a+b)}$

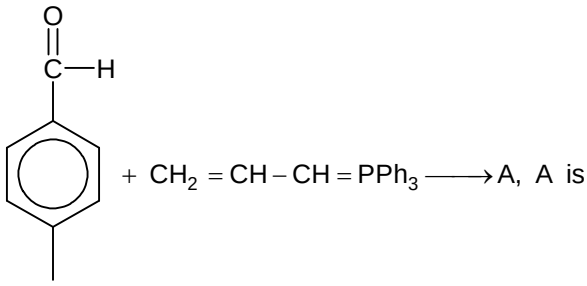
(B) $E_3 = \frac{aE_1^0 - bE_2^0}{(a+b)}$

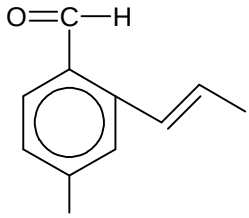
(C) $E_3 = \frac{bE_2^0 - aE_1^0}{(a+b)}$

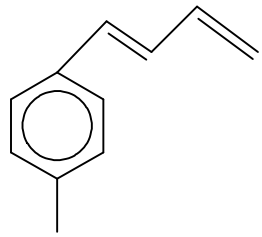
(D) $E_3 = \frac{aE_1^0 - bE_2^0}{(a-b)}$

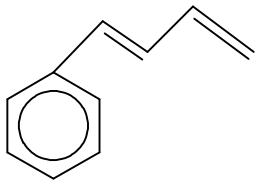
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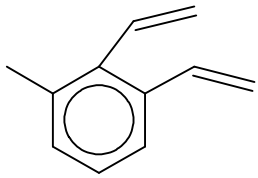
19. An inorganic molecule X on heating gives green colouration and evolve O_2 gas the X is
(A) $K_2Cr(SO_4)_2$ (B) $K_2Cr_2O_7$
(C) $RbCrO_4$ (D) CrO_2Cl_2

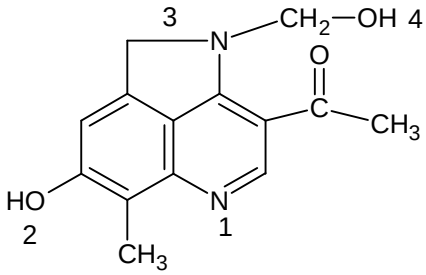
20. 
$+ \ CH_2 = CH - CH = PPh_3 \longrightarrow A$, A is

(A) 

(B) 

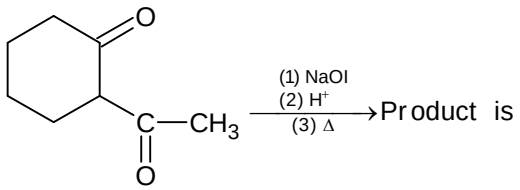
(C) 

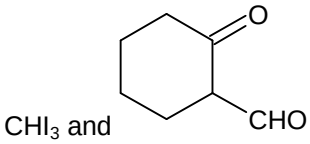
(D) 

21. 

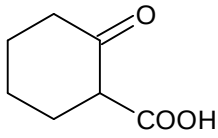
H^+ will attack on the basic site
(A) 1 (B) 2
(C) 3 (D) 4

Space for rough work

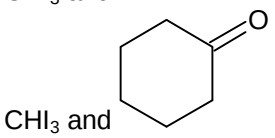
22. 

(A) 

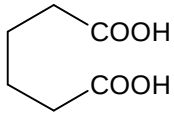
CHI_3 and

(B) 

CHI_3 and

(C) 

CHI_3 and

(D) 

CHI_3 and

23. Which of the following order is incorrect?
(A) $Na_2O < K_2O < Rb_2O$ (basic nature)
(B) $CH_4 > SiH_4 > GeH_4 > SnH_4$ (stability of hydride)
(C) $NH_3 < PH_3 < AsH_3$ (basic nature)
(D) $N_2O_5 < P_2O_5 < As_2O_5$ (acidic nature)

24. Select correct statement:
(A) Red oxide of lead is known as massicot
(B) Red oxide of lead is known as litharge
(C) Yellow oxide of lead is known as litharge
(D) Yellow oxide of lead is known as anarge

25. Most oxidizing agent is
(A) $\left[CeO_4\right]^{-2}$
(B) $\left[WO_4\right]^{-2}$
(C) $\left[CrO_4\right]^{-2}$
(D) $\left[MnO_4\right]^{-2}$

26. $\left[NiCl_4\left(NH_3\right)_2\right]^{-2}$ can show:
(A) Coordination isomerism
(B) Linkage isomerism
(C) Geometrical isomerism
(D) Hydration isomerism

Space for rough work

27. $As_2O_3 + H_2O \longrightarrow AsO_4^{-3} + H^+$, in the reaction n factor of As_2O_3 is

(A) 1 (B) 2
(C) 3 (D) 4

28. The mean free path (λ) of a gas sample is given by:

(A) $\lambda = \sqrt{2}\pi\sigma^2 N$ (B) $\lambda = \frac{1}{\sqrt{2}\pi\sigma^2 N}$

(C) $\lambda = \sqrt{2}\pi 4\sigma^2 N$ (D) $\lambda = \frac{\sqrt{2}}{\pi\sigma^2 N}$

29. The oxide of Alkali earth metal gives brick red coloured carbide with the formula. (M_2C), on reaction with carbon the metal oxide is:

(A) MgO (B) CaO
(C) BeO (D) SrO

30. Which of the following pair of metal form nitrite on reaction with Nitrogen?

(A) Li, Mg (B) Mg, Na
(C) Al, K (D) Al, Na

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Let $f(x) = \begin{cases} |x| & \text{for } 0 < |x| \le 3 \\ 1 & \text{for } x = 0 \end{cases}$, then at x = 0 f(x) has
   (A) a local maximum (B) no local maximum
   (C) a local minimum (D) no extremum

2. If three equations are consistent
   $(a + 1)^3x + (a + 2)^3y = (a + 3)^3$
   $(a + 1)x + (a + 2)y = a + 3$
   $x + y = 1$, then a is equal to
   (A) 1 (B) 2
   (C) -2 (D) 3

3. The remainder on dividing $1234^{567} + 89^{1011}$ by 12 is
   (A) 1 (B) 5
   (C) 9 (D) none of these

4. If tan A, tan B are the roots of the quadratic equation $abx^2 - c^2x + ab = 0$ where a, b, c are sides of the ΔABC then $\sin^2 A + \sin^2 B + \sin^2 C$ is
   (A) 1 (B) 2
   (C) 3 (D) $\frac{3}{2}$

5. If $I_1 = \int_{-100}^{101} \frac{dx}{(5 + 2x - 2x^2)(1 + e^{2-4x})}$ and $I_2 = \int_{-100}^{101} \frac{dx}{5 + 2x - 2x^2}$, then $\frac{I_1}{I_2}$ is
   (A) $\frac{1}{2}$ (B) 2
   (C) 0 (D) none of these

Space for rough work

6. Let f: R $\to$ R and g: R $\to$ R be two one-one and onto functions such that they are the mirror images of each other about the line y = 2. If h(x) = f(x) + g(x) then h(0) equals to
(A) 2 (B) 4
(C) 0 (D) none of these

7. If the straight line 3x + 4y = 24 intersects the axes at A and B and the straight line 4x + 3y = 24 intersects the axis at C and D then the number of parabolas possible to draw through A, B, C and D is
(A) 0 (B) 1
(C) 2 (D) 4

8. Two given circles of radii r_1 and r_2 touch each other externally. If $\frac{r_1}{r_2} = 3 + 2\sqrt{2}$ then the locus of the centre of a circle that touches both the circles externally is
(A) a straight line (B) a pair of perpendicular straight lines
(C) either of two parabolas (D) a rectangular hyperbola

9. In an equilateral triangle, inradius (r), circumradius (R) and exradius (r_1) are in
(A) A.P. (B) G.P.
(C) H.P. (D) none of these

10. $\lim\limits_{x\to\infty} \int\limits_0^{x^2/2} \frac{t^2}{x^2(1+t^2)} dt$ is equal to
(A) $\frac{1}{4}$ (B) $\frac{1}{2}$
(C) 1 (D) none of these

11. The number of pairs (m, n) of integers such that $n^2 - 3mn + m - n = 0$ is
(A) 0 (B) 1
(C) 2 (D) none of these

12. The greatest possible number of points of intersection of 6 straight lines and 4 circles is
(A) 55 (B) 75
(C) 45 (D) 51

Space for rough work

13. If $\cos 3x + \sin\left(2x - \frac{7\pi}{6}\right) = -2$, then x is equal to (where $m \in I$)

(A) $\frac{\pi}{3}(6m-1)$ (B) $\frac{\pi}{3}(6m+1)$

(C) $\frac{\pi}{3}(2m+1)$ (D) none of these

14. If a, b, c > 0, then the minimum value of $(a + b + c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$ must be

(A) 6 (B) 8

(C) 9 (D) 12

15. The value of $\int \frac{(x^2+1)}{(x^4-x^2+1)\cot^{-1}\left(x-\frac{1}{x}\right)}dx$ will be

(A) $\ln\left|\cot^{-1}\left(x-\frac{1}{x}\right)\right|+c$ (B) $-\ln\left|\cot^{-1}\left(x-\frac{1}{x}\right)\right|+c$

(C) $\ln\left|\cot^{-1}\left(x+\frac{1}{x}\right)\right|+c$ (D) none of these

16. The highest power of 3 contained in (70)! Must be equal to

(A) 16 (B) 24

(C) 32 (D) 48

17. If f(x) = sinx + cosax is periodic then a is

(A) 2 (B) π

(C) $\frac{\pi}{2}$ (D) $\sqrt{2}$

18. If A is a nilpotent matrix of index 2, then for any positive integer n, $A(I + A)^n$ is equal to

(A) A^{-1} (B) A

(C) A^n (D) I_n

Space for rough work

19. If in triangle ABC, $\angle C = 45°$ then value of $\sin^2 A + \sin^2 B$ lies in the interval
(A) [0, 1] (B) (0, 1)
(C) $\left(\frac{1}{2}, \frac{1+\sqrt{2}}{\sqrt{2}}\right)$ (D) none of these

20. The curve $xy = c$ $(c > 0)$ and the circle $x^2 + y^2 = 1$ touch at two points, then distance between the points of contacts is
(A) 1 (B) 2
(C) $2\sqrt{2}$ (D) none of these

21. A seven digit number without repetition and divisible by 9 is to be formed by using 7 digits out of 1, 2, 3, 4, 5, 6, 7, 8, 9. The number of ways in which this can be done is;
(A) 9! (B) 2 (7!)
(C) 4 (7!) (D) none of these

22. Point of intersection of the lines $\text{Arg}(z - 1) = \frac{\pi}{4}$ and $z = it + (1 - t)$ is
(A) i (B) 1
(C) 1 + i (D) none of these

23. If p and q be the longest and the shortest distances respectively of the point (–7, 2) from any point (α, β) on the curve whose equation is $x^2 + y^2 - 10x - 14y - 51 = 0$ then G.M of p and q is
(A) $2\sqrt{11}$ (B) $5\sqrt{5}$
(C) 13 (D) none of these

24. If the sides of a right angled triangle are in G.P then the cosines of the acute angles of the triangle are
(A) $\frac{\sqrt{5}-1}{2}, \frac{\sqrt{\sqrt{5}-1}}{2}$ (B) $\frac{-\sqrt{5}-1}{2}, \frac{\sqrt{\sqrt{5}-1}}{2}$
(C) $\frac{1}{2}, \frac{1}{4}$ (D) none of these

25. The number of real root(s) of the equation $x^2 \tan x = 1$ lie(s) between 0 and 2π is/are
(A) 1 (B) 2
(C) 3 (D) 4

Space for rough work

26. If α, β and γ are the roots of the equation $x^2(px + q) = r(x + 1)$. Then the value of determinant $\begin{vmatrix} 1+\alpha & 1 & 1 \\ 1 & 1+\beta & 1 \\ 1 & 1 & 1+\gamma \end{vmatrix}$ is

(A) $\alpha\beta\gamma$ (B) $1+\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}$

(C) 0 (D) none of these

27. If $f(x) = x^3 + ax^2 + bx + c$ has local maxima at certain $x \in R^+$ and minima at certain $x \in R^-$ then

(A) $b > 0, c > 0$ (B) $b > 0, c < 0$

(C) $b < 0$ (D) none of these

28. The number of solutions of $\log_e|x| = \sqrt{2-x^2}$ is

(A) 4 (B) 1

(C) 2 (D) 3

29. The value of $c\int_{1+c}^{a+c}[f(cx)+1]dx - \int_{c}^{ac} f(c^2+x)dx$ $(c \neq 0)$, is equal to

(A) 0 (B) $c(a - 1)$

(C) ac (D) $a(c + 1)$

30. If z_1, z_2, z_3 are unimodular complex numbers then the greatest value of $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$ equals to

(A) 3 (B) 6

(C) 9 (D) $\frac{27}{2}$

Space for rough work